Exercise 4: Joins, UNION, filtering and aggregates.

Brightlearn
Nyandoro Nathoma

1. SQL JOINS

1. INNER JOIN

```
SELECT EmployeeID,
       firstName,
       Lastname,
       Department,
       Salary,
       ProjectID,
       Project Name,
       Budget,
       Status
FROM Employees AS A
INNER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	firstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100 000	Completed
1	John	Doe	IT	70000	103	Cyber security Audit	75 000	Pending
2	Alice	Smith	HR	60 000	102	Employee training	50 000	ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90 000	ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	65 000	Completed
6	Michael	clark	Finance	80 000	106	Risk Management	80 000	Pending

Exercise 4: JOINS, filtering and aggregates

BrightTeam
Nyandano Nathomas

2. LEFT JOIN

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       ProjectID,
       ProjectName,
       Budget,
       Status
FROM Employees AS A
LEFT JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70 000	101	AI development	100 000	Completed
1	John	Doe	IT	70 000	103	Cyber Security Audit	75000	Pending
2	Alice	Smith	HR	60 000	102	Employee Training	50 000	Ongoing
3	Bob	Johnson	finance	75 000	104	Financial Analysis	90 000	Ongoing
4	David	Brown	IT	72 000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65 000	105	Market Expansion.	65000	Completed
6	Michael	Clark	finance	80 000	106	Risk Management	80 000	Pending.

Exercise 4: JOINS, filtering and aggregates

Brightlearn
Nyarolano Nathoma

3. RIGHT JOIN

```
SELECT projectID,
       Project Name,
       Budget,
       Status,
       EmployeeID,
       firstName,
       LastName,
       Department,
       Salary
FROM projects AS A
RIGHT JOIN Employees AS B
ON A.EmployeeID = B.EmployeeID;
```

ProjectID	Project Name	Budget	Status	Employee ID	firstName	LastName	Department	Salary
101	AI derelopment	100 000	Completed	1	John	Doe	IT	70 000
102	Employee Training	50 000	ongoing	2	Alice	Smith	HR	60 000
103	Cybersecurity Audit	75 000	Pending	1	John	Doe	IT	70 000
104	financial Analysis	90000	ongoing	3	Bob	Johnson	finance	75 000
105	Market Expansion	65 000	Completed	5	Emma	Wilson	Sales	65 000
106	Risk Management	80 000	Pending	6	Michael	Clark	finance	80 000

Exercise 4: JOINS, filtering and aggregates

Brightlearn
Nyandarua Wathoma

4. FULL OUTER JOIN

```
SELECT EmployeeID,
       FirstName,
       Last Name,
       Department,
       Salary,
       ProjectID,
       Project Name,
       Budget,
       Status
FROM Employees AS A
FULL OUTER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	Project Name	Budget	Status.
1	John	Doe	IT	70 000	101	AI development	100 000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75 000	Pending
2	Alice	Smith	HR	60 000	102	Employee Training	50 000	Ongoing
3	Bob	Johnson	finance	75000	104	Financial Analysts	90 000	Ongoing
4	David	Brown	IT	72 000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	finance	80 000	106	Risk Management	80 000	Pending

Exercise 4: JOINS, UNION, filtering and aggregates

BrightLearn
Nyandano Mathoma

3. filtering Statements

7. ~~Emplo~~

```
SELECT EmployeeID,
       firstName,
       LastName,
       Department,
       Salary
FROM employees
WHERE Salary>70 000;
```

Employees Salary More than 70000

EmployeeID	firstName	LastName	Department	Salary
3	Bob	Johnson	finance	75 000
4	David	Brown	IT	72000
6	Michael	Clark	finance	80 000

8.

```
SELECT EmployeeID,
       firstName,
       Last Name,
       Department,
       Salary
FROM Employees
WHERE Department = 'IT'
  OR Department = 'finance';
```

Employees in IT or finance

EmployeeID	firstName	LastName	Department	Salary
1	John	Doe	IT	70 000
3	Bob	Johnson	finance	75000
4.	David	Brown	IT	72 000
6	Michael	Clark	finance	80 000

9.

```
SELECT ProjectID,
       ProjectName,
       Budget,
       Status
FROM Projects
WHERE Status <> 'Completed';
```

Projects Not Completed.

ProjectID	ProjectName	Budget	Status.
102	Employee training	50000	Ongoing
103	Cybersecurity Audit	75 000	Pending
104	Financial Analysis.	90 000	Ongoing
106	Risk Management	80 000	Pending.

Exercise 4: JOINS, filtering and aggregates. Projects with: Bright learn Nyandaro Mathoma

10.

```
SELECT ProjectID,
       ProjectName,
       Budget,
       Status
FROM projects
WHERE Budget > 70000
AND Status <> 'Completed';
```

Budget greater than 70000 and not completed

ProjectID	ProjectName	Budget	Status.
	Cybersecurity Audit	75 000	Pending.
	Financial Analysis	90 000	Ongoing
	Risk Management	80 000	Pending

11.

```
SELECT EmployeeID,
       FirstName,
       Last Name,
       Department,
       Salary,
       City
FROM Employees
WHERE City = 'New York'
OR City = 'Toronto'
ORDER BY Salary DESC;
```

Employees from New York or toronto ordered by Salary DESC

EmployeeID	FirstName	LastName	Department	Salary	City
6	Michael	Clark	finance	80000	New York
3	Bob	Johnson	finance	75000	Toronto
1	John	Doe	IT	70000	New York

12.

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary
FROM Employees
ORDER BY Salary DESC,
LIMIT 3;
```

Top 3 Highest paid employees.

EmployeeID	FirstName	LastName	Department	Salary
6	Michael	Clark	finance	80 000
3	Bob	Johnson	finance	75000
4	David	Brown	IT	72 000

Exercise 4: JOINS, filtering and aggregates

BrightHearn Nyandoro Mathoma

4. Aggregate functions With ~~Grou~~ GROUP BY & HAVING

13.

```
SELECT Department,
       SUM(Salary) AS TotalSalary
FROM Employees
GROUP BY Department
ORDER BY TotalSalary DESC;
```

Total Salary Per department

Department	TotalSalary
finance	155 000
IT	142 000
Sales	65 000
HR	60 000

14.

```
SELECT city,
       AVG(Salary) AS AverageSalary
FROM employees
GROUP BY City
HAVING AVG(Salary) > 65000;
```

Average Salary Per City > 65000

City	AverageSalary
New York	75 000
Toronto	75 000
London	72000

15.

```
SELECT Department,
       COUNT(EmployeeID) AS EmployeeCount
FROM Employees
GROUP BY Department
HAVING COUNT(EmployeeID) > 1;
```

Employees per department (>1)

Department	EmployeeCount
IT	2
finance	2

16.

```
SELECT status,
       COUNT(ProjectID) AS ProjectCount
FROM Projects
GROUP BY status
HAVING COUNT(ProjectID) ≥ 2;
```

Number of Projects per status

Status	ProjectCount
Completed	2
Ongoing	2
Pending	2

Exercise 4: JOINS, filtering and aggregates

Brightlearn
Nyandano Nathoma

17. Total Project Budget per employee which is greater than 150 000

```
SELECT EmployeeID,
       firstName,
       Last Name,
       ~~SUM(Budget) AS Total Project B~~
       SUM(Budget) As TotalProjectBudget
FROM Employees As A
    JOIN Projects AS B
   ON A.EmployeeID = B.EmployeeID
GROUP BY EmployeeID,
         first Name,
         Last Name
HAVING sum(Budget) > 150 000;
```

EmployeeID	firstName	LastName	TotalProjectBudget
1	John	Doe	175 000